UM Hackathon 2026

Z.AI YTL*AI LABS

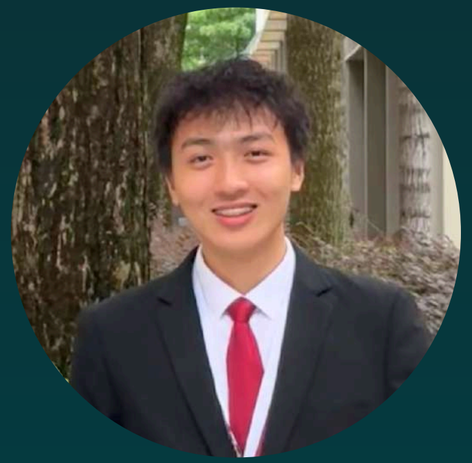
Team ID : 247

Team Awawa

DOMAIN 1 :
AI SYSTEMS & AGENTIC WORKFLOW AUTOMATION

Our Team

Awawa



Ooi Rui Zhe

Undergraduate Year 3
Artificial Intelligence
University Malaya

Vanness Liu

Undergraduate Year 3
Artificial Intelligence
University Malaya

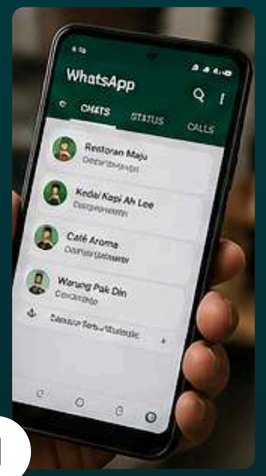
Ong Zhao Qian

Undergraduate Year 3
Artificial Intelligence
University Malaya

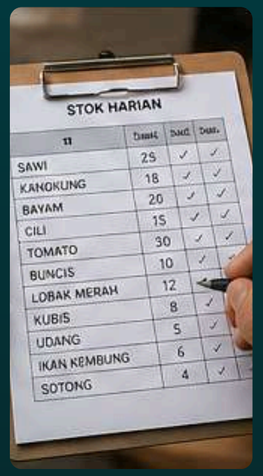
User Story

Pak Ahmad,
Grocery Supply Business Owner,
Klang, Selangor

EVERYDAY TASKS

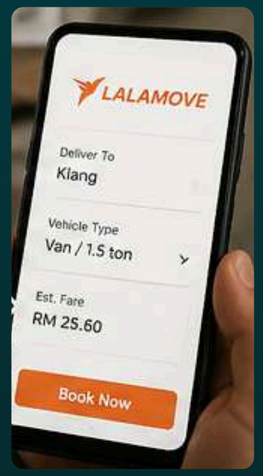


1 Read WhatsApp manually



2 Cross-check Stock Manually

3 Calculate Pricing by Hand



4 Book logistic manually

> *Most of my time goes into handling orders, not growing the business.*

Background

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Majority of Malaysia's Wholesale Supply Chain Runs on WhatsApp

The Reality in Malaysia's B2B Trade

Default Infrastructure on WhatsApp

SME wholesalers manage hundreds of daily orders ***entirely through social messaging apps***.

Mixed Language Complexity

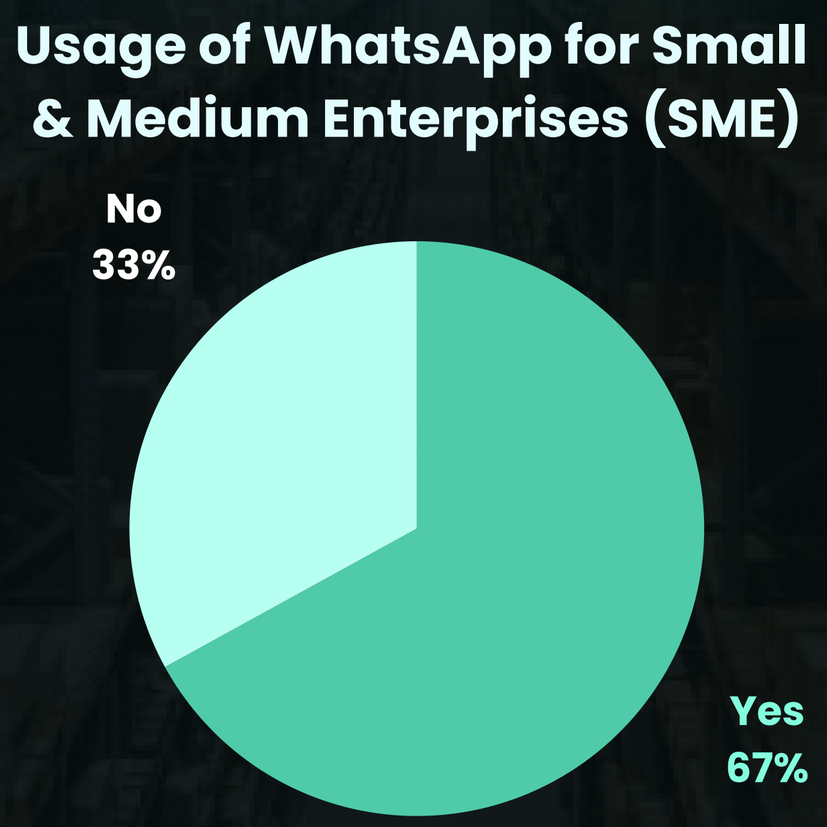
Orders arrive as rapid-fire text, voice notes and calls in ***Bahasa Rojak and abbreviations***.

Habit of Shadow Workflows

Behind every chat is an ***invisible manual process***: staff cross-referencing on clipboards, whiteboards, manually calculating discounts and re-typing addresses in logistics booking.

Problem Statement

Heavy Reliance on WhatsApp for Business Communication



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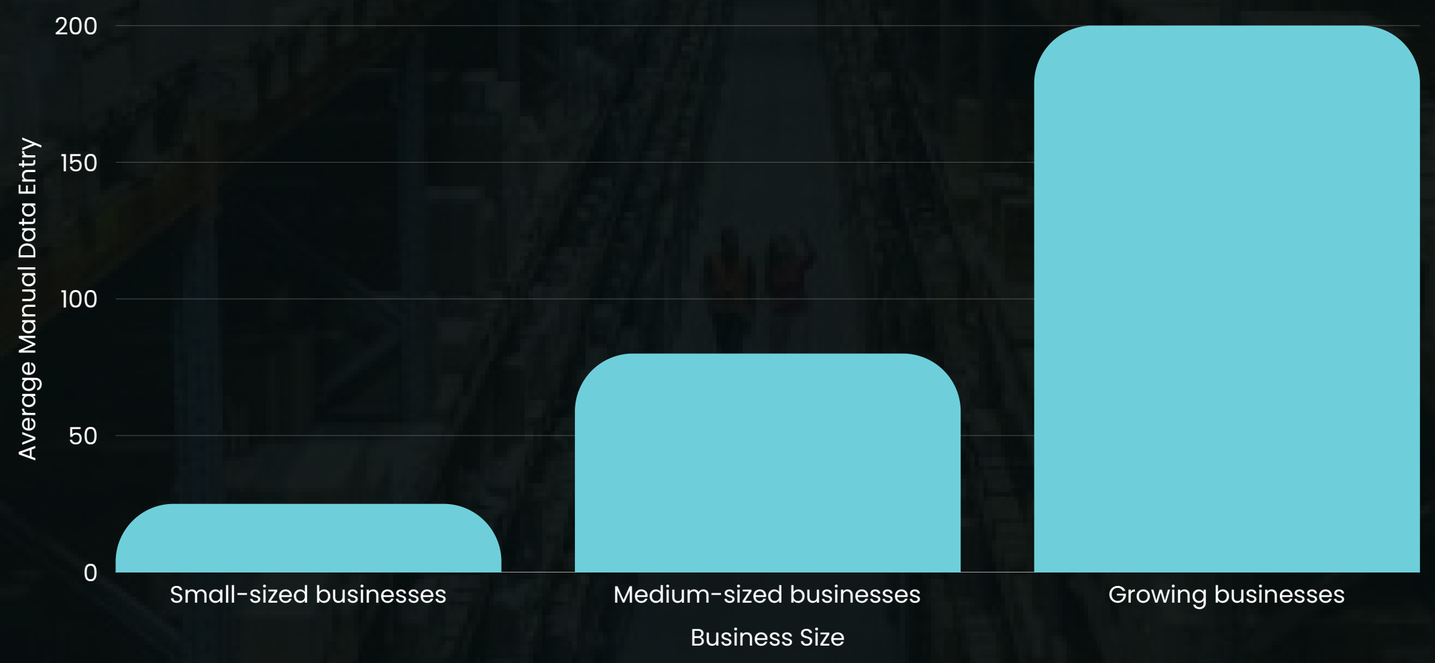
Source: DOSM's 2024 SME Performance Report

Massive Effort to Scale Unstructured Messages Daily



Source: The Hidden Cost of Manual Data Entry for SMBs - XCloudConnector

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Costly Margin of Human Error

Manual data entry by non-specialist employees results in a 5% to 10% error rate

Source: Social Media Usage among Small Enterprises during Covid-19 Pandemic - HM Publishers

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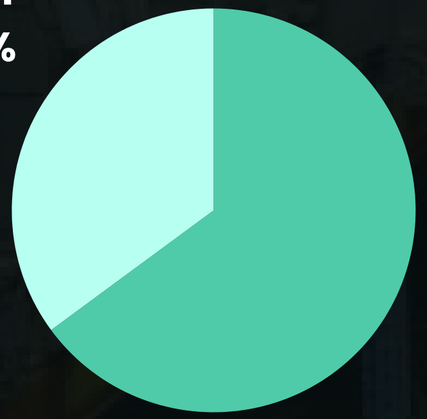
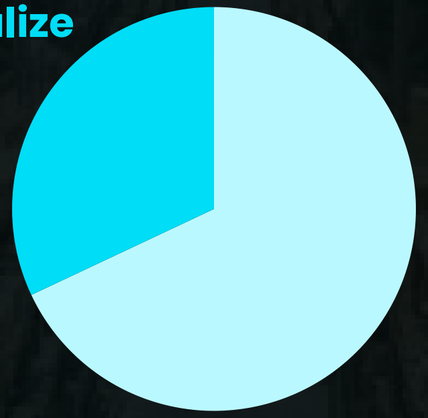
Massive Market Opportunity and Digital Gap

Wholesale & Retail Contribution to MSME Sector



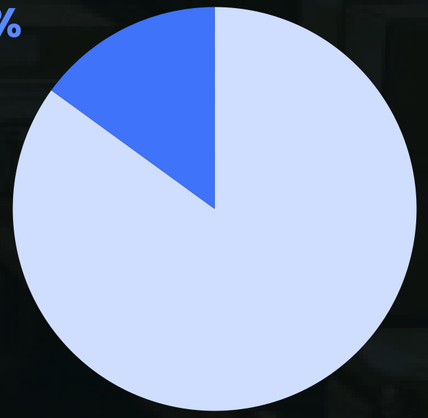
[Big Market Opportunity]

SME Digitalization Adoption



[Low Digitalization]

AI Adoption in SMEs



[Low AI Adoption]

Source: Dynamic Capabilities and Industry 4.0 Adoption, SME Corp targets the digitalisation of all SMEs by 2024, & ENHANCING PERFORMANCE IN SMES

Problem Statement

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Summary

Heavy Reliance on WhatsApp for Business Communication

Massive Effort to Scale Unstructured Messages Daily

The Costly Margin of Human Error

Market Opportunity and Digital Gap

Target Users

- Wholesale Suppliers
- SME Distributors
- Retail Buyers
- Warehouse Managers

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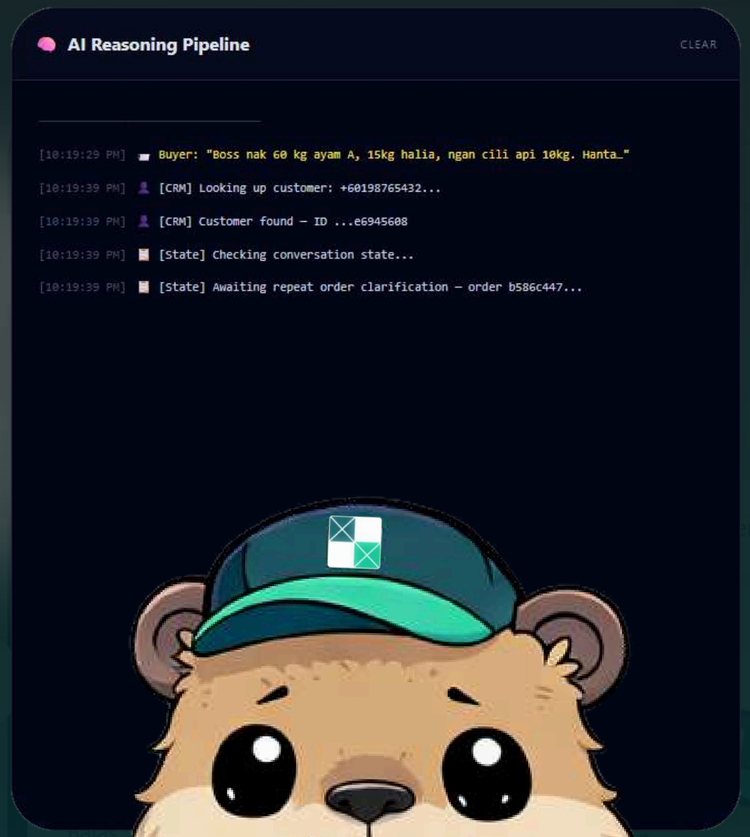
8 DECENT WORK AND ECONOMIC GROWTH

12 RESPONSIBLE CONSUMPTION AND PRODUCTION



SupplyLah

From Chat. To Chain.

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What SupplyLah wants to achieve?

Automate Conversational Order Intake

Scale Daily Message Processing Operations via Multi-Agent AI

Eliminate Operational Errors & Coordination Burnout

Bridge the SME Digitalization Gap

SupplyLah

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SupplyLah

Key Features of SupplyLah

- WhatsApp Message & Voice Intake
- Live Inventory Read
- WhatsApp Order Confirmation to Buyer
- Voice Notes Transcription to Order
- Customer Confirmation Loop
- Order Log and Basic Reporting
- GLM-Powered Multilingual Order Parsing and Inventory Reasoning
- Inventory Deduction on Order Confirmation
- Logistics Booking via External API
- Wholesaler Staff Operations Dashboard and Alert Notification

Techstack Architecture

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App Service Plan

Backend API

FastAPI
FastAPI + Python

MiniLM Embeddings

Docker

AWS EC2

Frontend Web App

Next.js
React

Vercel
Vercel Edge Network

Database & Storage

Supabase

AWS S3

State & Queue Management

Redis

External API Services

Large Language Models

OpenAI Whisper
Groq OpenAI Whisper

Gemini 2.5 Flash

Payment API

Touch 'n Go eWallet (Simulated Payment API)

Communication & Telephony API

Twilio WhatsApp Business API

Resend Transactional Email

Logistics API

Lalamove API (Simulated Logistic API)

Implementation Flow

SupplyLah

1. Receive client message
2. Debounce Buffer
3. Modality Detection
4. Audio Pre-Processing
5. Speech Transcription (Groq Whisper-v3)
6. Image OCR (Gemini 2.5 Flash)
7. Prompt Construction (Intake Agent)
8. Order Extraction via Gemini 2.5 Flash (Intake Agent)

9
Confidence Gate (≥80%)

10
Repeated Failure Escalation

11
Human Review & Resume

12
Inventory & Rules Query (Inventory Agent)

13
Stock Feasibility Evaluation (Inventory Agent)

14
Substitution Proposal

15
Order Confirmation & Timeout

16
Logistics Booking & Dispatch (Logistics Agent)

SupplyLah
From Chat. To Chain.

16-step autonomous pipeline. 3 AI agents.
Zero manual order entry.

AI-POWERED WHOLESALE

Turn WhatsApp Chaos Into Automated Orders

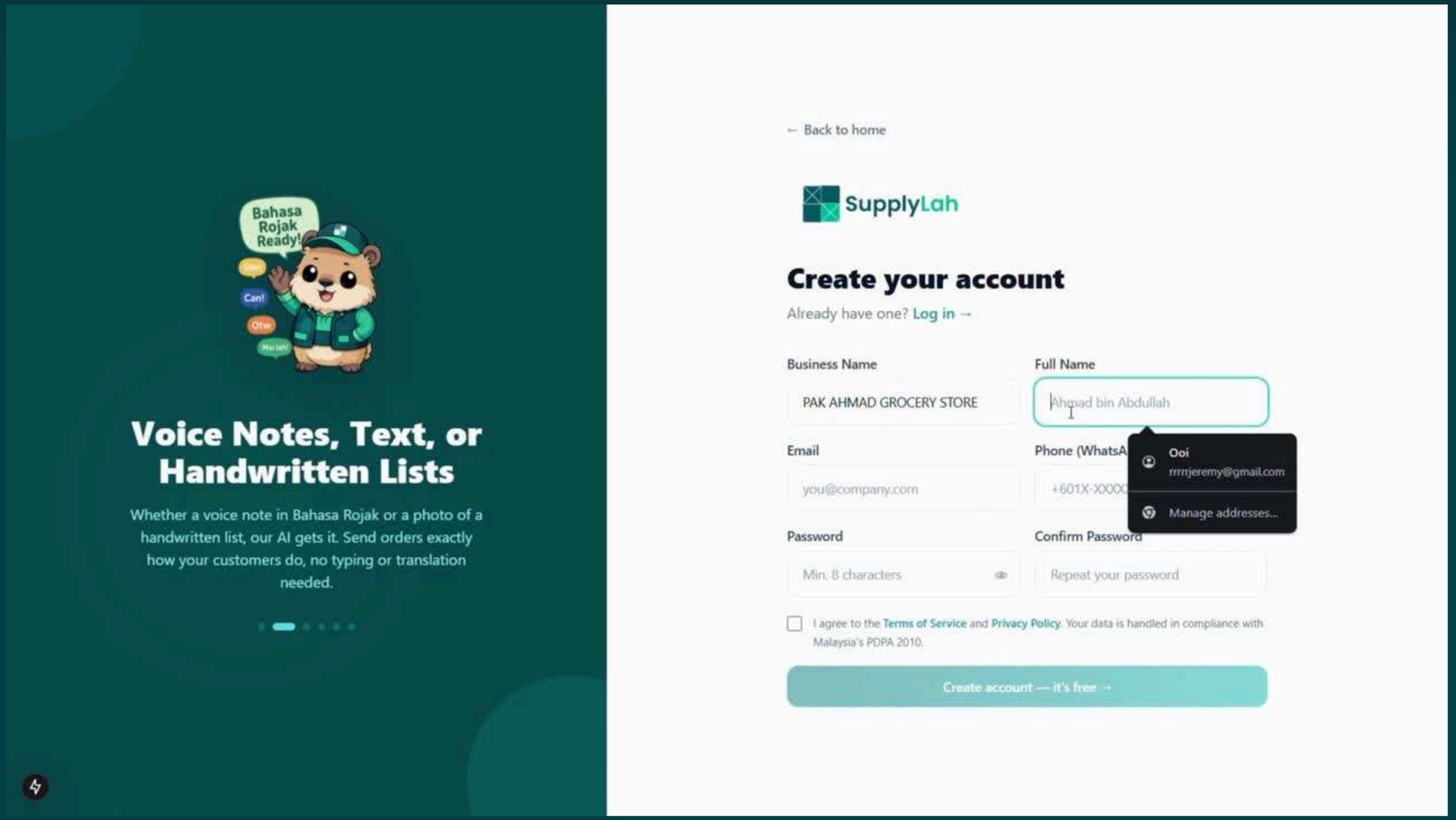
Built for real wholesale workflows, SupplyLah converts **messy multilingual WhatsApp orders** into **ready-to-fulfill, structured orders**, and automates **stock verification, pricing, and delivery**, reducing manual work to near zero.

Get Started Free

View Live Demo

Project Demo

Account Sign Up



Supabase

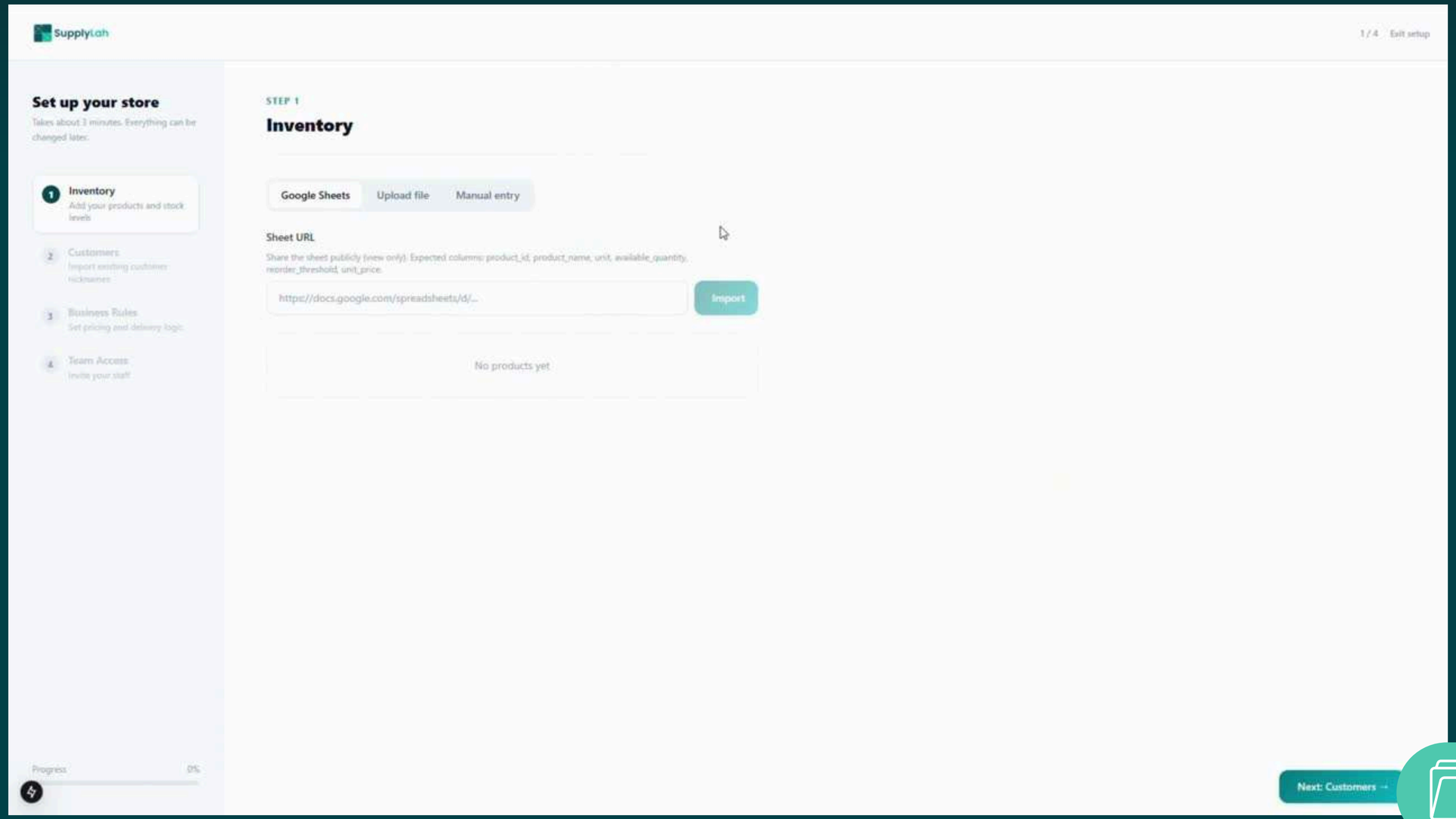
- Auth
- Database
- RLS

Resend

- Confirm Signup

Project Demo

Account Setup (Get started)



Google Sheet

Excel

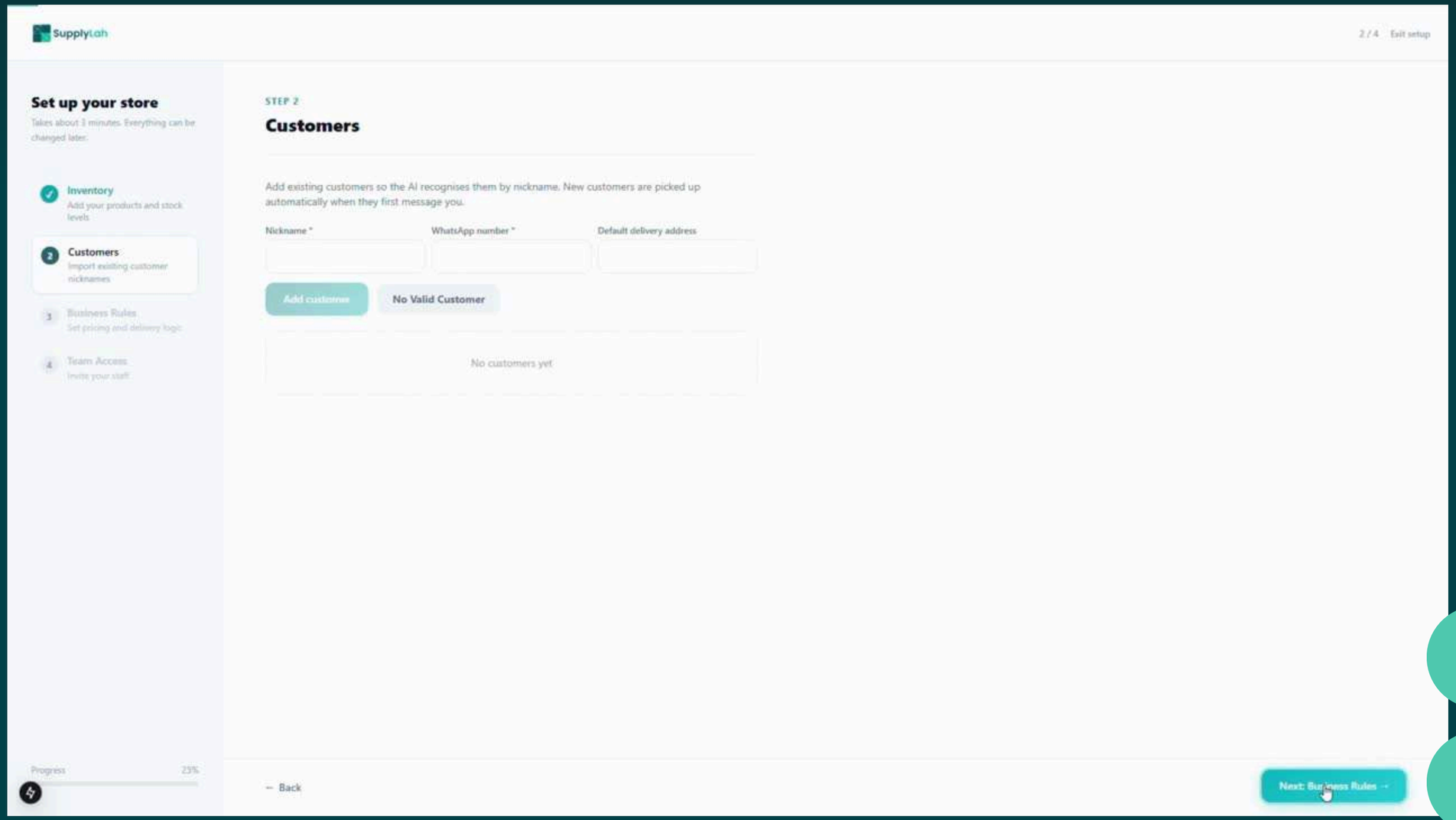
CSV

Manual Entry

Import Inventory data

Project Demo

Account Setup (Get started)

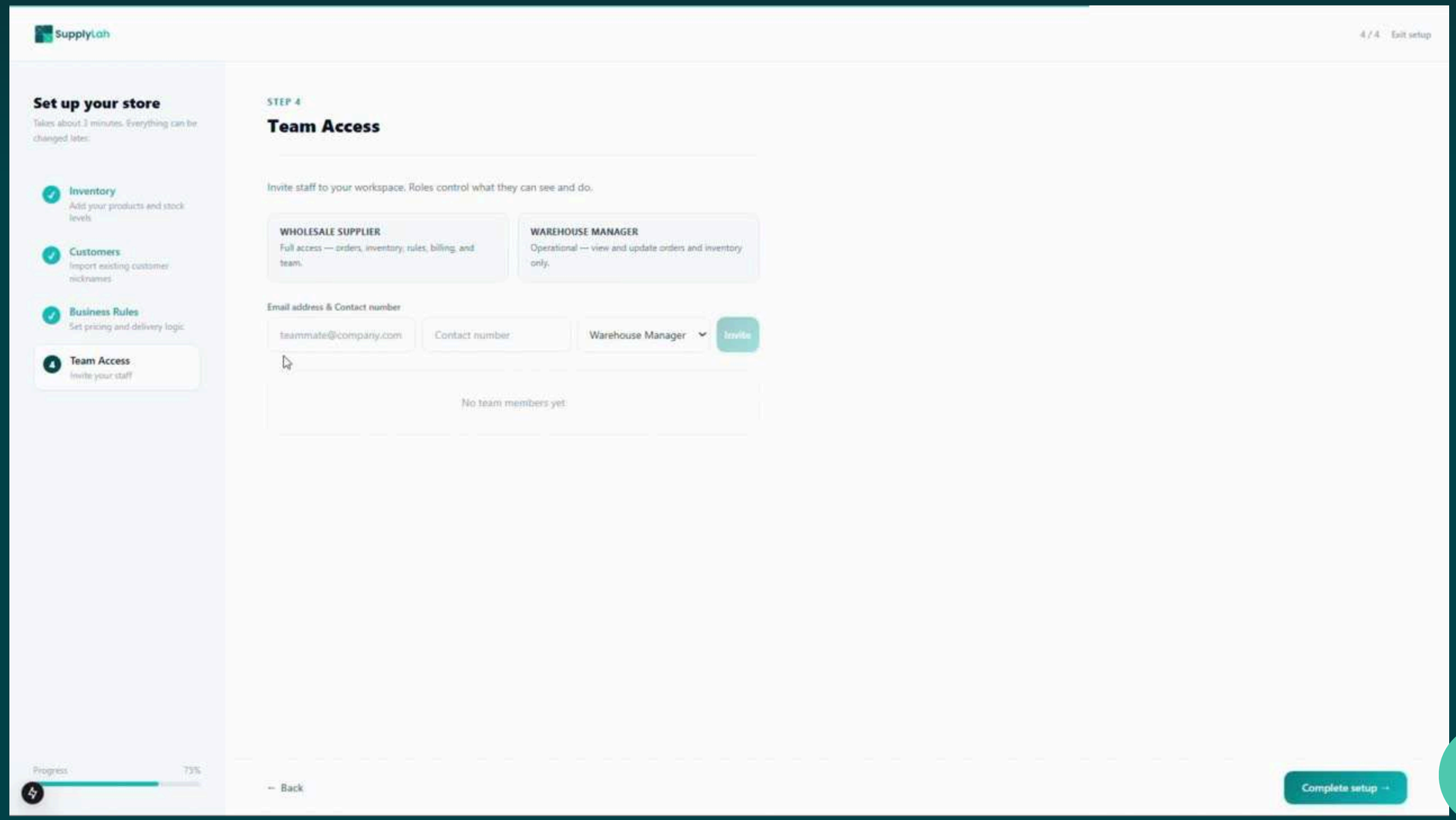


Enter Customer Data

Setup Business Rule

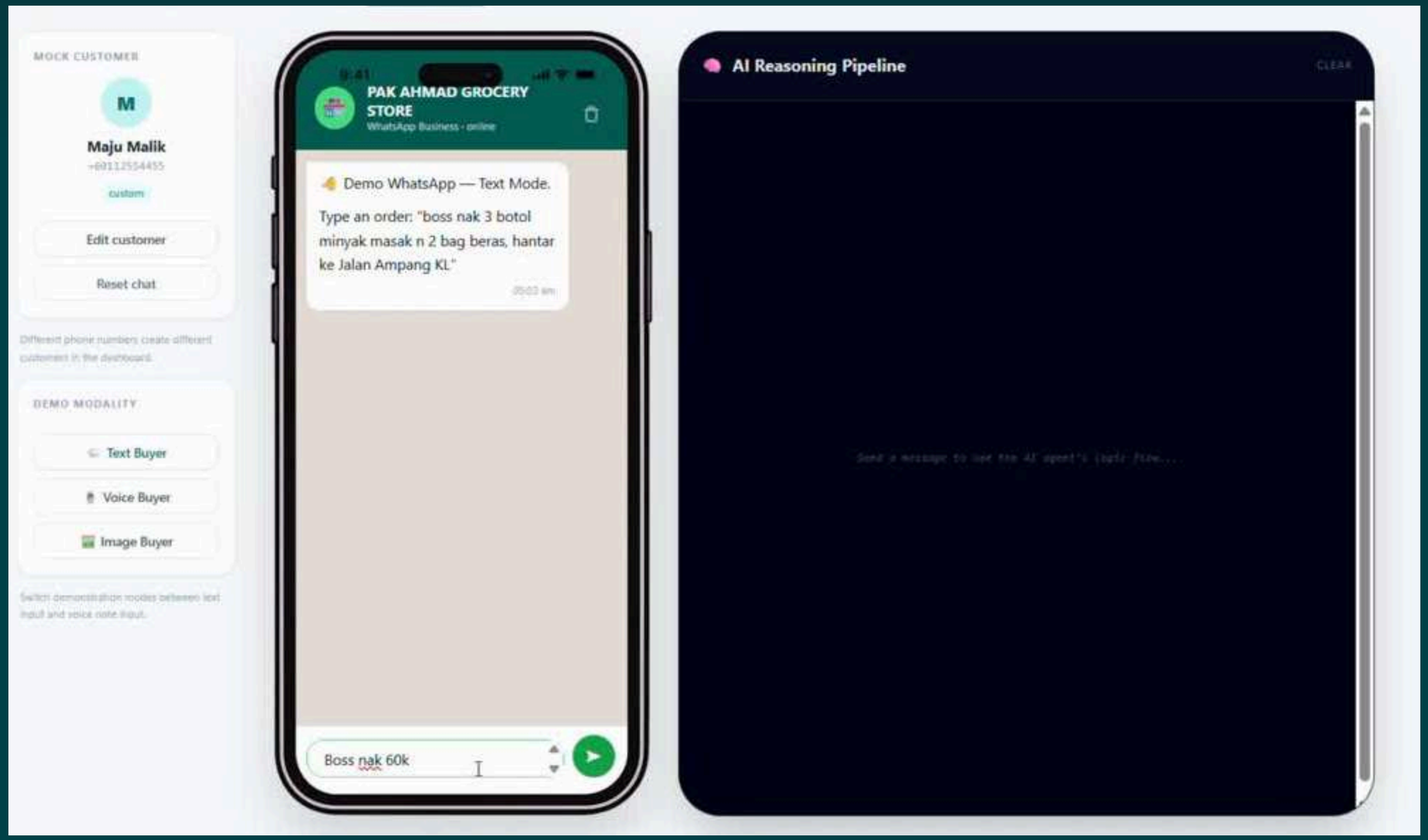
Project Demo

Account Setup (Get started)



Team Invitation

Project Demo

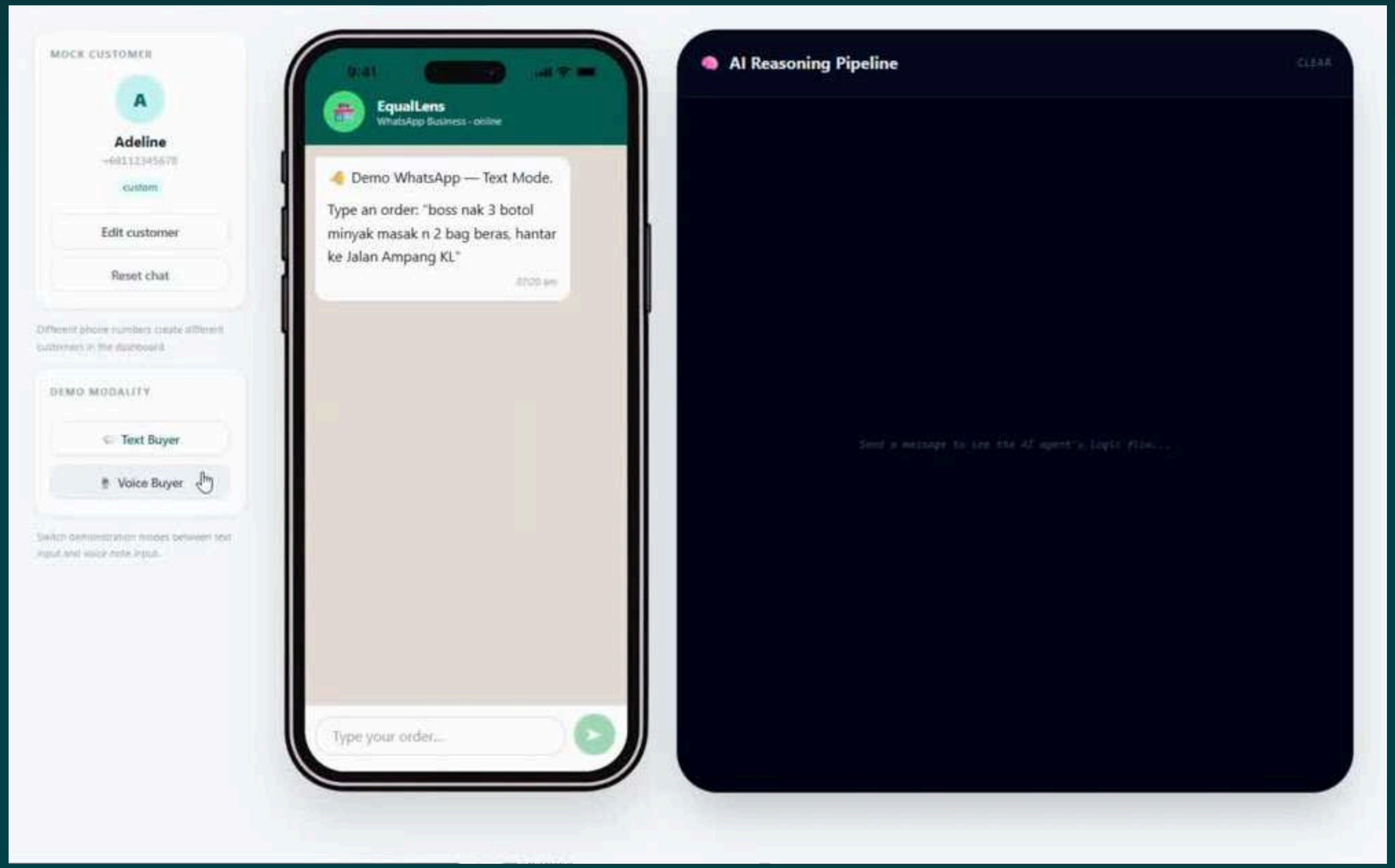


Exp Text Order Intake

- 3.1 Send unstructured text order
- 3.2 System Acknowledgement
- 3.3 Intake Agent — NLP Parsing
- 3.4 Hybrid Product Resolver
- 3.5 Live Stock Check
- 3.6 Order Confirmation
- 3.7 Payment Gating
- 3.8 Atomic Stock Deduction
- 3.9 Logistics Booking
- 3.10 Delivery Confirmation

Project Demo

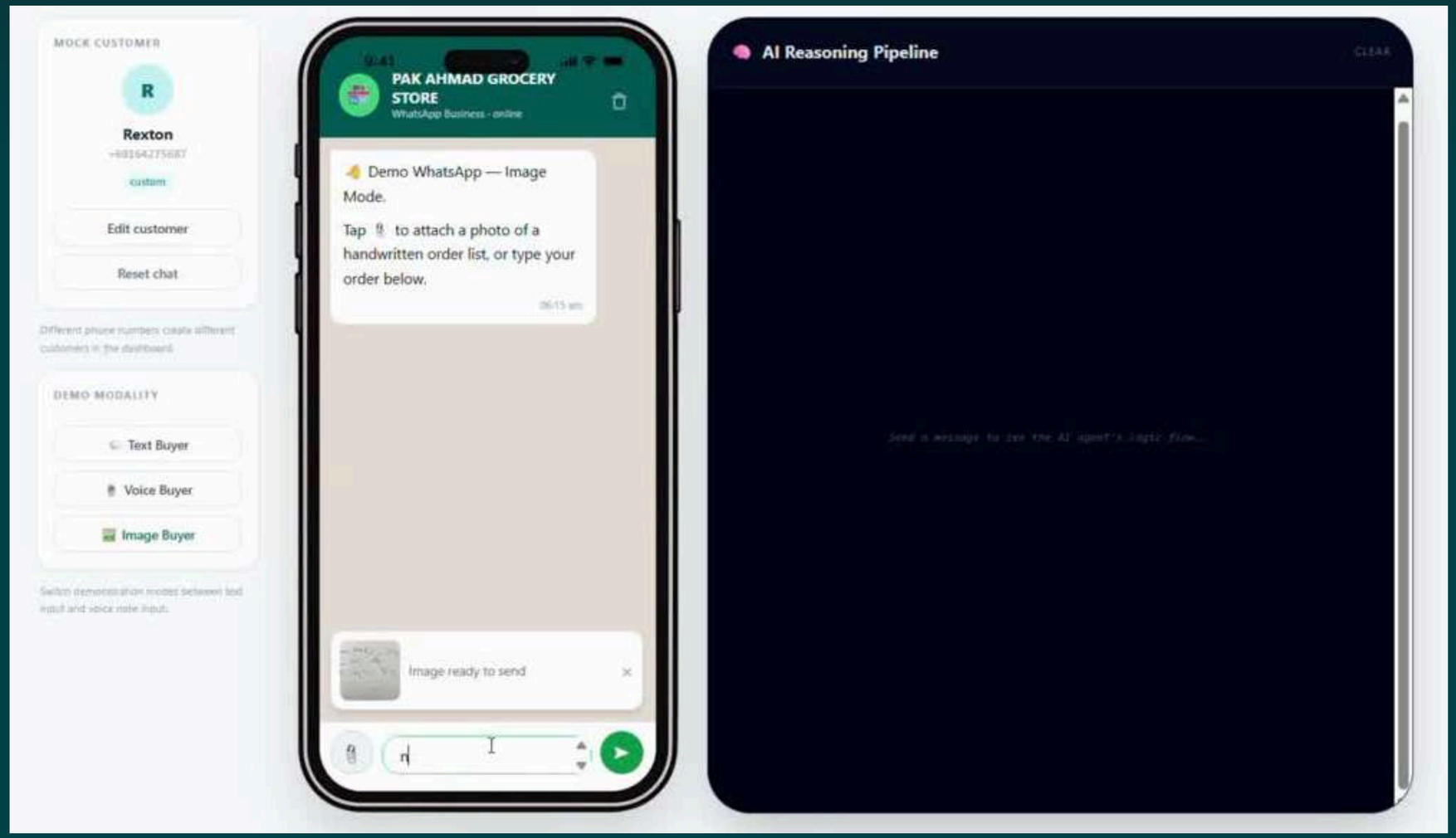
Exp Voice Order Intake



- 4.1 Send Voice Note Order
- 4.2 Audio Pre-Processing
- 4.3 Audio Transcription
- 4.4 System Acknowledgement
- 4.5 Intake Agent — NLP Parsing
- 4.6 Delivery Address Checking
- 4.7 Buyer Sends Address
- 4.8 Live Stock Check
- 4.9 Substitution Logic Triggered
- 4.10 Substitution Proposal Sent to Buyer

Project Demo

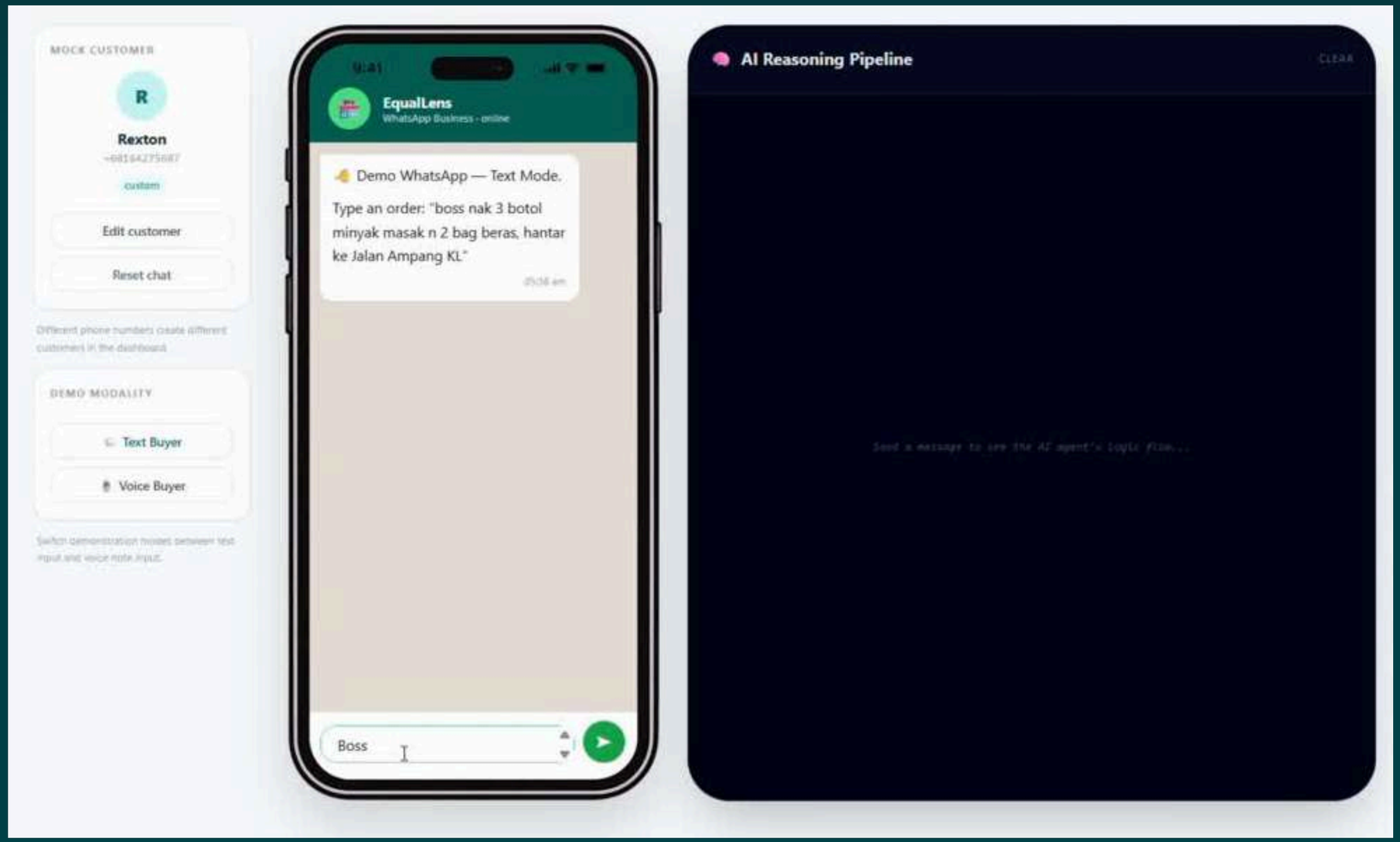
Exp Image Order Intake



- 5.1 Send Image Order List
- 5.2 Extract Text from Image
- 5.3 System Acknowledgement
- 5.4 Intake Agent — NLP Parsing
- 5.5 Hybrid Product Resolver
- 5.6 Delivery Address Checking
- 5.7 Buyer Sends Address
- 5.8 Live Stock Check

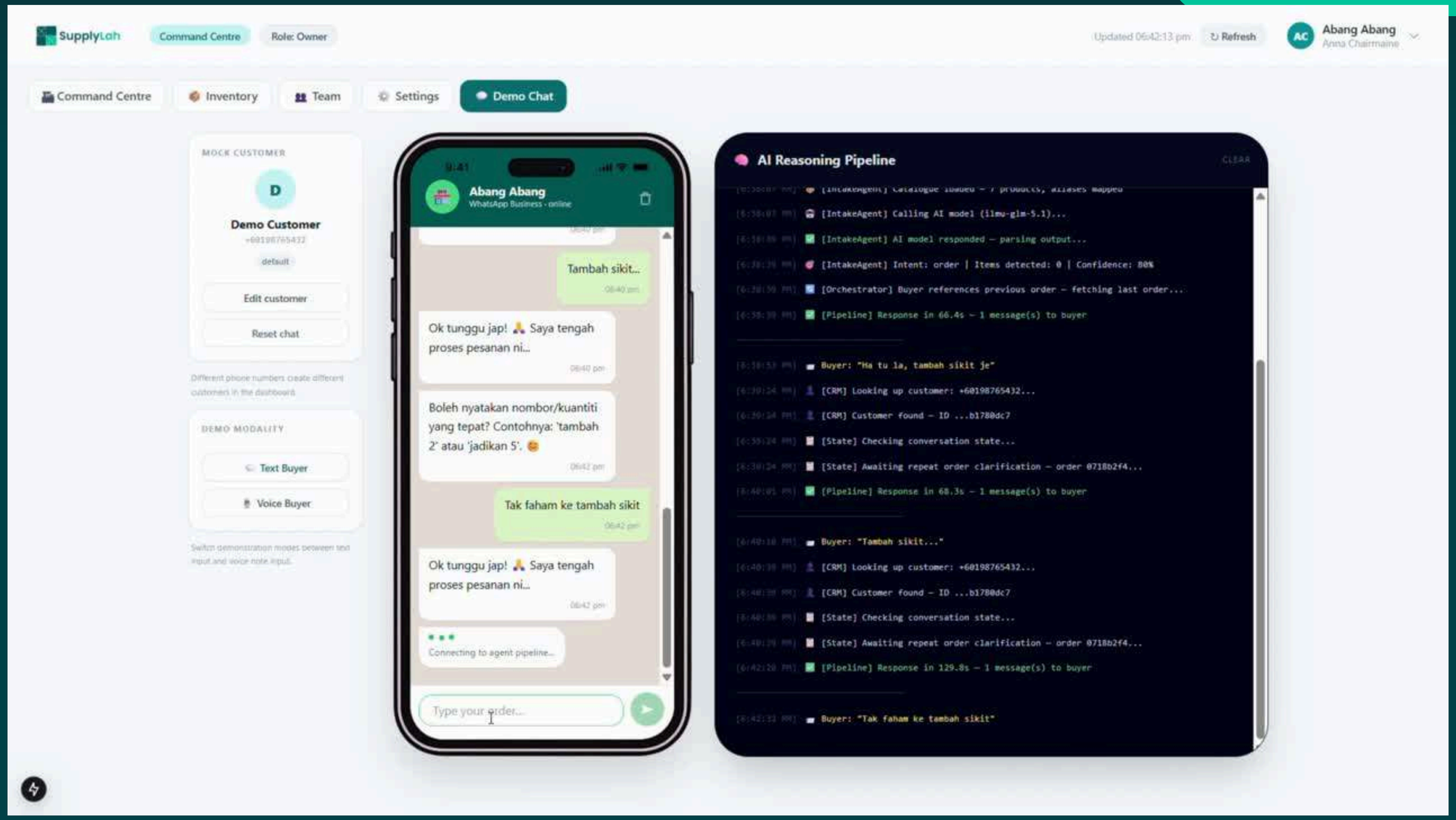
Project Demo

Ambiguous Order and Low Confidence Parsers



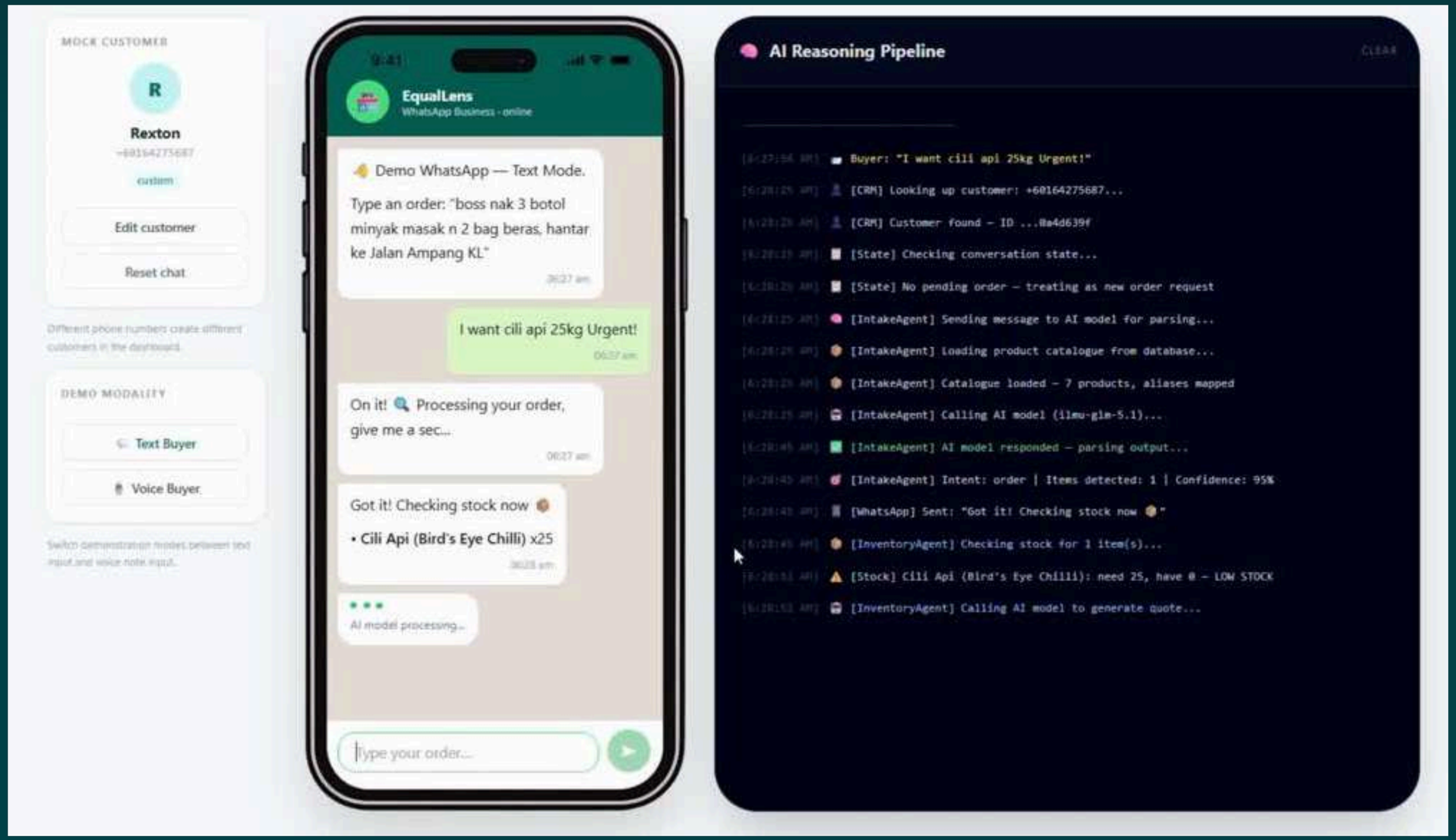
Project Demo

Repeated Low Confidence Escalation to Staff



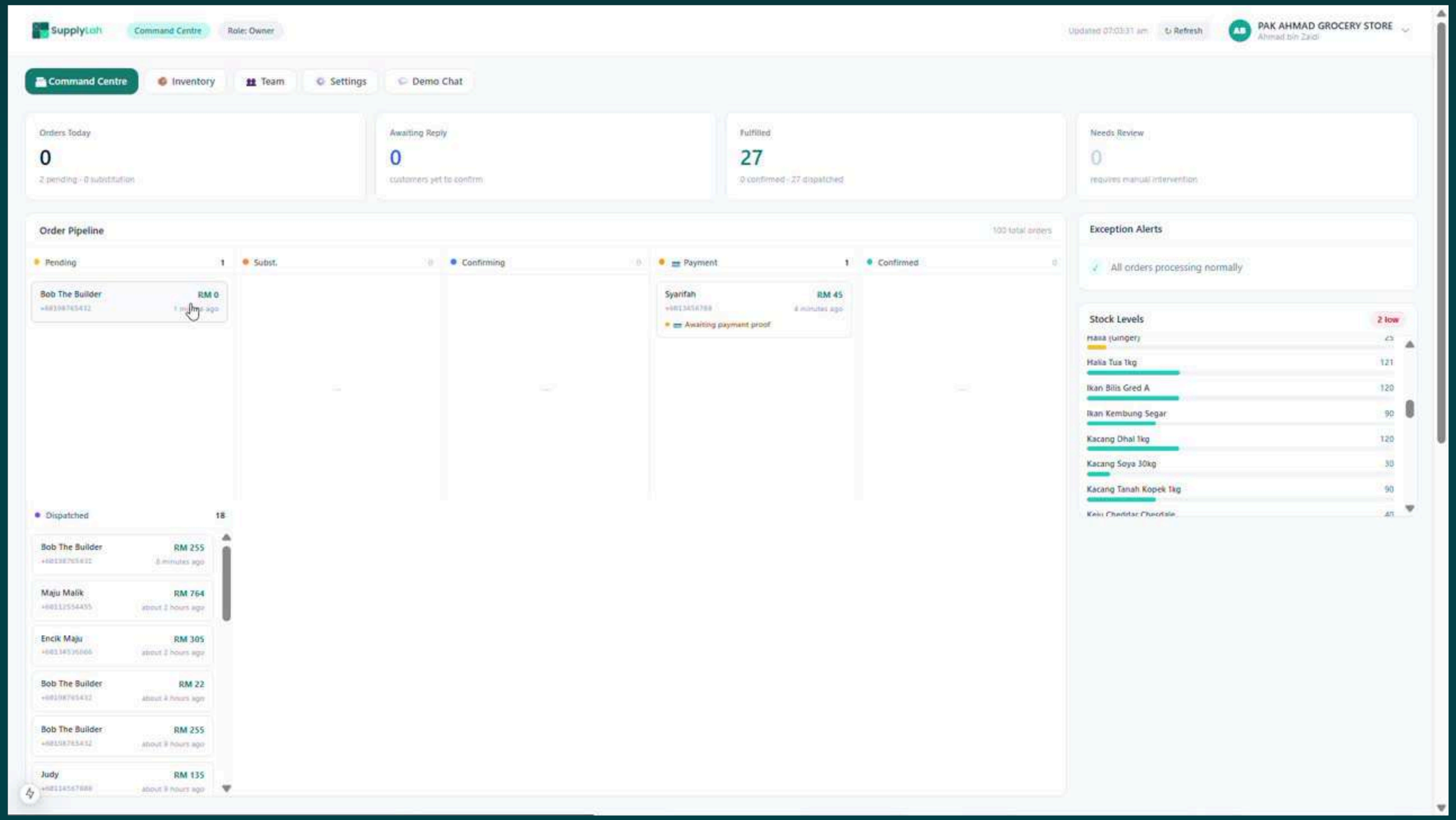
Project Demo

Out of Stock and No Substitution Available

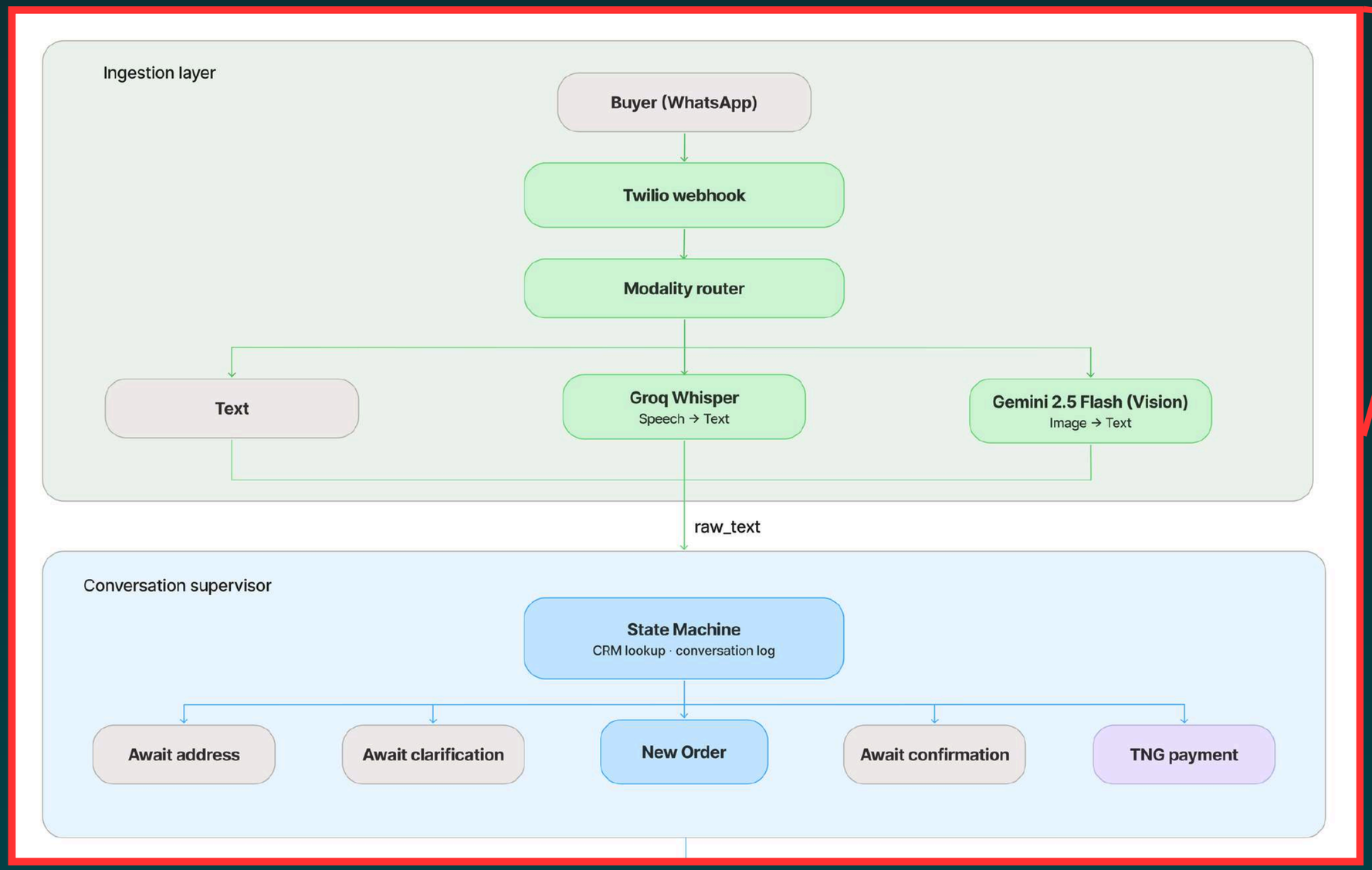


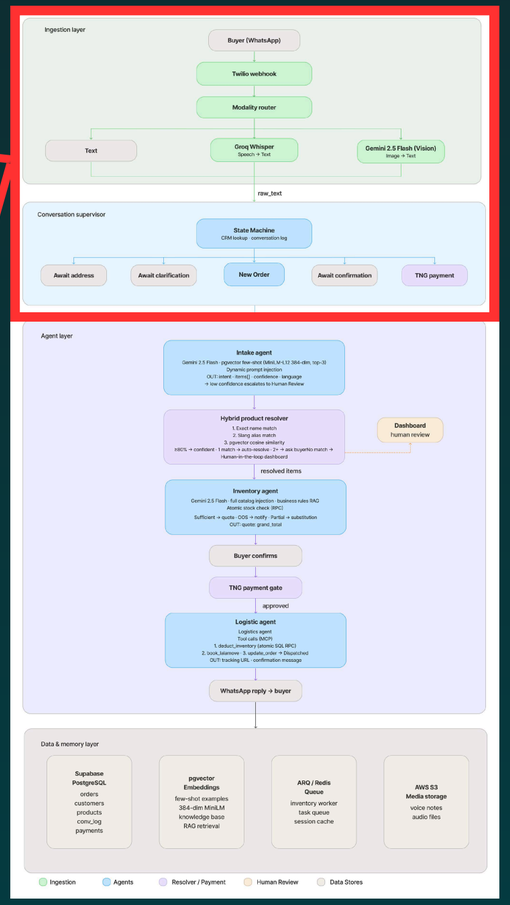
Project Demo

Live Order Kanban

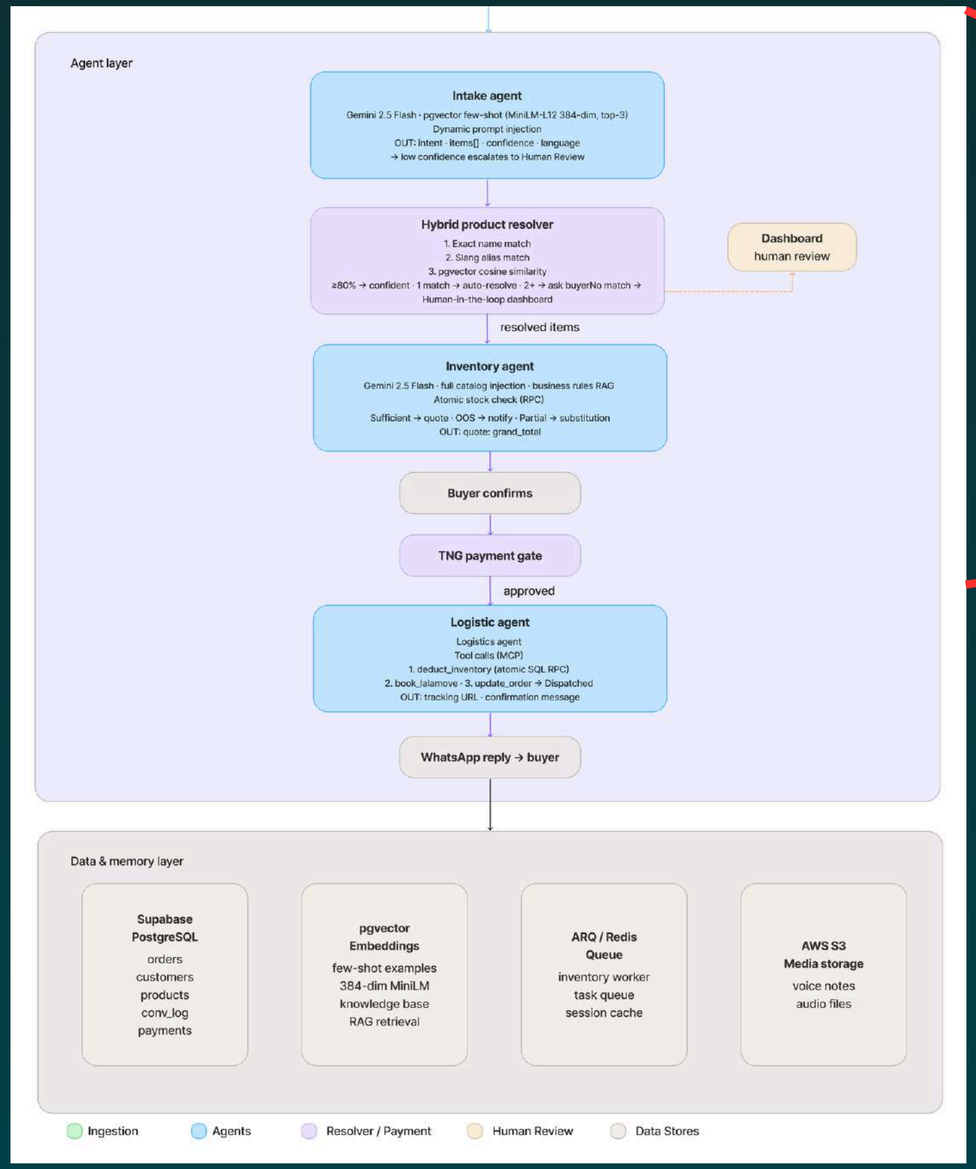


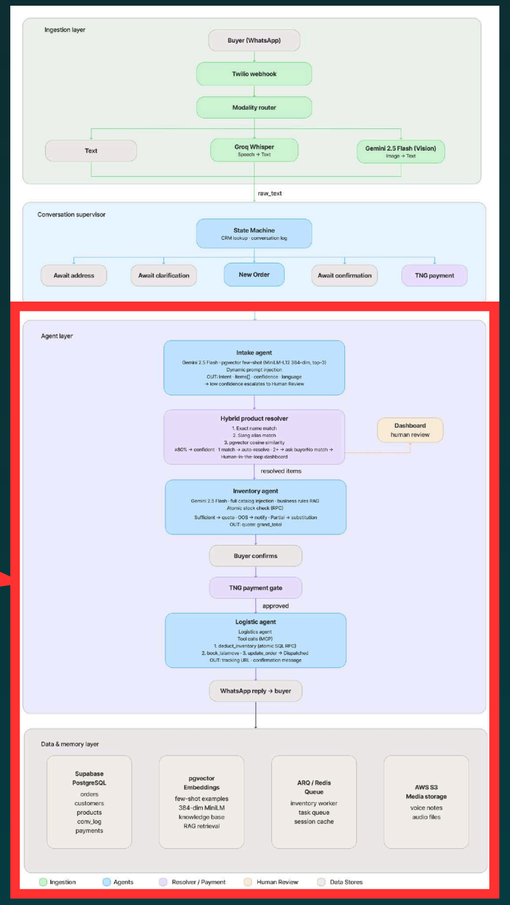
Agentic Order Processing Pipeline





Agentic Order Processing Pipeline





Production-Grade AI & System Reliability

NLP Parsing of Intake Agent

Using pgvector semantic search, 3 dynamic few-shot examples is compared with buyer's message before parsing by Gemini. Low confidence will be flagged for human review before handling the

3-Stage Product Resolver

Exact name match, product alias match and lastly, pgvector cosine similarity are used. Product options are presented to buyer for clarification when similarity score is ambiguous.

Hallucination Guardrails

LLM clarification message are overriden by templates, inventory agent retries up to 3 times on malformed JSON responses and substitutions always require YES/NO explicitly from buyer.

Atomic Stock Deduction

PostgreSQL RPC executes single conditional updates for deducting stock, eliminating race conditions and prevent overselling when concurrent orders occur.

SupplyLah

SupplyLah

Deployment

Technical Stack & Scalability

- Dockerized FastAPI (AWS) & Next.js Edge (Vercel)
- Supabase PostgreSQL with real-time WebSockets
- Async background workers for heavy AI tasks
- Auto-scaling containers & DB connection pooling

CI/CD Pipeline & Automation

- Fully automated GitHub Actions pipeline
- pytest quality gates for core AI logic
- Blue-Green deployments for zero downtime
- Dynamic feature toggles (Hot-swap Mock to Live APIs)

Integration Security & Credentials

- Zero hardcoded API keys (Strict .gitignore)
- Secrets injected via encrypted Environment Vaults
- Row-Level Security (RLS) for strict tenant data isolation
- TLS 1.3 encryption & secure webhook authentication

Test Execution & Automation Evidence

```
tests/test_core_logic.py::test_UT01_detect_bahasa_rojak
[UT-01] Input: 'Boss, nak hantar ayam' -> Actual Result: 'ms'
PASSED
tests/test_core_logic.py::test_UT02_detect_english
[UT-02] Input: 'Please deliver tomorrow' -> Actual Result: 'en'
PASSED
tests/test_core_logic.py::test_UT03_confirm_affirmative
[UT-03] Input: 'Ok lah' -> Actual Result: True
PASSED
tests/test_core_logic.py::test_UT04_confirm_negative_slang
[UT-04] Input: 'tak nak' -> Actual Result: False
PASSED
```

Automated Unit Tests

- **Tool Used:** Pytest (Isolated Logic)
- **Tested:** "Bahasa Rojak" parsing (e.g., identifying "tak nak" as negative)
- **Tested:** Pricing & substitution math (Zero hallucinations)

API & Integration Tests

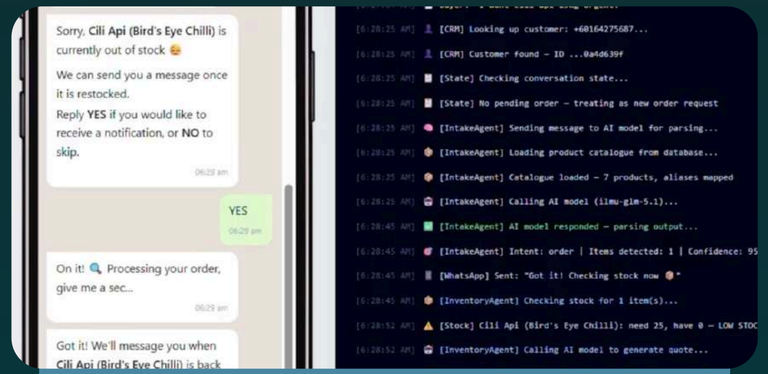
- **Tested:** Webhook → Supabase data persistence
- **Tested:** Stateful context memory (Pending → Confirmed)
- **Tested:** Real-time WebSocket UI updates

```
{
  "from_number": "+60123456789",
  "text_content": "Boss, hantar 50kg ayam ke Selayang",
  "message_type": "text",
  "merchant_id": "test_merchant"
}
```

Body Cookies Headers 4 Test Results 200 OK 29.85 s

Raw Preview Visualize

1 OK



Edge Cases & UAT

- **Tested:** Extracting unstructured, multi-item intent
- **Tested:** Graceful halting on "Out of Stock" (No system crashes)
- **Tested:** Human-in-the-loop escalation alerts

Token Efficiency & Cost Management

Text & Image Order

Intake Agent	~3,550 (input)	+	~125 (output)	=	~3,675 tokens			
	~1,860 (ocr input)	+	~150 (ocr output)	+	~3,450 (text input)	+	~125 (text output)	= ~5,585 tokens
Inventory Agent	~4,240 (input)	+	~500 (output)	=	~4,740 tokens			
Logistics Agent	~725 (input)	+	~200 (output)	=	~925 tokens			
Total Cost	~13,825 (input) ($0.3/1M)	+	~1,100 (output) ($2.5/1M)	=	~$0.007			

Voice Order

Intake Agent	~3,590 (input)	+	~125 (output)	=	~3,715 tokens
Inventory Agent	~4,240 (input)	+	~500 (output)	=	~4,740 tokens
Logistics Agent	~725 (input)	+	~200 (output)	=	~925 tokens
Total Cost	~30 ($0.111/hr) +	~8,555 ($0.3/1M)	+	~825 ($2.5/1M)	= ~$0.0056

Blended Total

$$\frac{\sim\$0.007 + \sim\$0.0056}{3} = \sim\$0.0042$$

$2.1 or RM 8.3 / mth

500 orders estimate

No Round-Trips

Both intake & inventory agents inject full product catalogue into the context, without it, it needs 5 separate Gemini calls for intake alone (initial parsing, tool call, stock check, tool result, retry).

Thus, 1 Gemini call per Agent

Context Caching

pgvector semantic search retrieves 3 most-relevant few-shot examples.

Inventory prompt uses a shared system prompt which is suitable for Gemini context caching at about 800 tokens

Output is constrained to structured JSON to prevent bloat.

Market Size 6.0%

Total Addressable Market

70M

Wholesale Trade SMEs across Southeast Asia

TAM

Serviceable Available Market

500k

Malaysian Wholesale & Retail SME Sectors

SAM

Serviceable Obtainable Market

30k

Tech-Ready SMEs Wholesalers in Malaysia

SOM

Source:

Resilient and Inclusive ASEAN Development of MSME in Asean, 2026 | Economic Census 2023: Wholesale and Retail Trade, 2023

Go-to-Market Strategy

SupplyLah

Phase 1

Hub-and-Spoke Rollout

Months 1 - 12

Target

Soft launch is done focusing on Klang Valley wholesale hubs, onboarding trial wholesalers

Goal

Reach 30 active paying operator clients by earning trust and validation through R&D and feature expansion

Phase 2

Regional Corridor Expansion

Months 13 - 24

Target

Rollout to Northern and Southern corridors across Peninsular Malaysia.

Goal

Expand aggressively to reach 120 active operator clients, reaching

Industry Vertical Plot

Direct partnerships with trade associations to onboard members from fresh produce, poultry & wet market communities.

Digital Clerk Trial Program

Free 60 days to first 50 wholesalers to prove its immediate ROI and reduce administrative overhead.

Buyer-Side Virality

Zero-learning-curve drives natural adoption as buyer demands fast, structured supplier service.

Business Model Canvas

Key Partners

- **Technology Partners:** Google Gemini, Groq, Supabase, AWS
- **Communication Partners:** Twilio WhatsApp Business API
- **Logistic Partners:** Lalamove, GrabExpress
- **Distribution Partners:** Malaysian wholesale trade associations

Key Activities

- AI Unstructured Order Parsing
- Real-time Inventory Synchronisation
- Prompt Engineering & New Feature Development
- Hub-and-Spoke Rollout

Key Resources

- **Proprietary Multi-Agent Tech Stack**
- **State-of-the-Art AI Models**
- **Cloud Infrastructure Architecture**

Value Propositions

1. **Zero Friction Adoption** where no new downloads are needed for buyers
2. **Multilingual Understanding** of parsing unstructured text, voice notes and images
3. **End-to-End Order Automation** handling order parsing, live stock verification, substitution negotiation and logistic booking
4. **Cost and Error Reduction** eliminating manual data entry staff mishaps
5. **Prevent Overselling** with atomic stock deduction

Customer Relationships

- Zero Learning Curve
- Free Trials of 60 days free order processing
- Human-in-the-Loop Support
- Easy Onboarding

Channels

- Industry Vertical Pilots
- Direct Operator Outreach
- Buyer-Side Virality
- User-friendly Platform Interface

Customer Segments

- **Primary Customers**
  - SME Wholesale Suppliers
  - Wholesale Distributors
- **Secondary Customers**
  - Retail Buyers
  - Restaurant Operators
  - Wet Market Vendors
- **Tertiary Customers**
  - Warehouse Managers
  - Operational Staffs

Cost Structure

- **Fixed Costs of Overhead & Maintenance:** Cloud infrastructure, R&D and feature development, sales & admin, onboarding support
- **Variable Costs Based on Usages:** AI token costs, messaging API fees, cloud database & media storage scaling

Revenue Stream

- Primary Source (**SaaS Subscription Plans** from RM 149 to RM 699 / month)
- **Pay-Per-Use Fees** of RM 0.15 fee per successful order
- Secondary Source (**Logistics Commission** of up to 5%)
- Tertiary Add-Ons (**Premium Analytics** at RM 150 per operator)

Competitive Advantages Analysis

Feature \ Brand	SupplyLah	STOREHUB	odoo	FOOD MARKET HUB	borong	wati	Manychat
WhatsApp-Native UX	✅	❌	❌	❌	❌	✅	✅
Inventory Management	✅	✅	✅	✅	✅	❌	❌
"Bahasa Rojak" Entity Extraction	✅	❌	❌	❌	❌	✅	✅
Business Rule Reasoning	✅	✅	✅	✅	✅	❌	❌
Non-Linear State Management	✅	❌	❌	❌	❌	❌	❌
Adaptive Negotiation	✅	❌	❌	❌	❌	❌	❌
Logistics Tool-Calling	✅	✅	✅	✅	✅	✅	✅
Autonomous Tool-Calling	✅	❌	❌	❌	❌	❌	❌
Stateful Order Lifecycle	✅	✅	✅	✅	✅	❌	❌

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Ongoing Key Features

- Multi-Channel Support
- Live In-App Payment Collection
- Live Voice Call Transcription to Order
- Dynamic Multi-Drop Logistics Management
- Driver Tracking & Live Map Feature
- Handwritten Order Image Parsing
- Customer ERP & Accounting Integration
- Full Business Intelligence Analytics
- Multi-Tenant SaaS Architecture

Future Roadmap

0-6 months

- Productionize Multimodal Input Parsing
- Integrate In-App Payment Collection with Local Payments
- Live Logistics API Transition to Lalamove & GrabExpress
- Drive Initiate Market Traction through Pilot Programs

6-12 months

- Develop Multi-Channel Support Beyond WhatsApp
- Implement Dynamic Multi-Drop Logistics Management
- Native Deployment Driver Track & Live Map Features
- Scale B2B Marketings and Door-to-door Onboarding

12+ months

- Customer ERP & Accounting Integration with Bespoke API Connectors
- Full Business Intelligence Analytics and Interactive Dashboards
- Transition to Multi-Tenant SaaS Architecture
- Regional Expansion across Peninsular Malaysia

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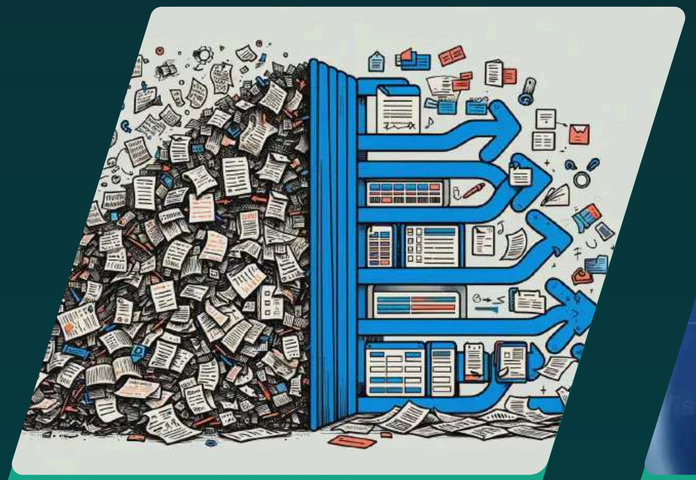
Bridging the Gap between Conversational Commerce & Formal Infrastructure



Alleviates Highly Unstructured Order Parsing Tasks

Reduce Human Errors and Redundancy

Equip SMEs with Tech Digitalisation Literacy

Thank you.

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From Chat. To Chain.